

Statistical Analysis of Running Performance Across Time of Day and Running Surface

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In modern society, we have a lot of shoe choices. This is the most considerable question when we are purchasing shoes. It is well known that whether or not a shoe fits can directly or indirectly affect a user's ability to exercise. Shoe design companies have practiced sports biomechanics research on various foot types (shape, length, walking habits...). The fit depends on the model used in making the shoes. In order to ensure that their products are competitive, shoe companies usually have their own unique models, which directly leads to the shoes produced by different companies with different products having different degrees of fit on people's feet. For the purpose of finding the best shoes for common people. Running is the most suitable method to test whether the shoes are fit or not since there would be a large friction between feet and shoes, hence the pros and cons of shoes would be enlarged during the running experiments. Therefore, we select the different brands of shoes and test them by 100 meter running in order to find the most suitable brand of shoes.

Factors

- Brand (3 brands) : levels Nike/Adidas/Converse
- Environment: straight road, uphill, grass field: levels
- Data Collection: (Time) Interested in the time (seconds) it takes to finish the 100m run.
- Group (3 runners): Blocking, aims to control variability in runner skill (speed, consistent, capabilities, shoe size) as a nuisance factor
- Blocking: environment, controlling the variability in weather condition
- Controlled nuisance factor: weather condition, runner physical health, time of day, shoe type/size, athletic abilities
- Uncontrolled nuisance factor: wind resistance, runners body temperature,

2c) Three basic principles of experiment

Blocking: Since the invited runners are varied in skill, capabilities, and shoes, I have chosen the runners as selected blocks.

Randomization: For each of the blocks, the order in which the nine-treatment combinations are run is randomly determined.

Replication: Each treatment combination is replicated three times, once in each block.

2d)

Present my design:

The experiment is designed by using three levels of time period factors and three levels of environment factors, and three runners are blocks. The experiment is performed by randomly selecting a treatment combination ('environment and time) within each block. therefore, we will follow a 3*3 factorial experiments run in a randomized complete block. the total design matrix will have a sample size of $3*3*3 = 27$ runs.

The statistical model for this experiment is :

$$y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \delta_k + \varepsilon_{ijk} \quad \left\{ \begin{array}{l} i = 1,2,3 \\ j = 1,2,3 \\ k = 1,2,3 \end{array} \right.$$

for $i = 1, 2, 3$, $j = 1, 2, 3$ and $k = 1, 2, 3$ where μ represents the overall mean, τ_i presents the time period effect, β_j presents environment effect, $(\tau\beta)_{ij}$ is the interaction, and δ_k is the block effect, and ε_{ijk} is the $N(0, \sigma^2)$ error component.

Data tables and descriptions :

<u>Test Subject Block:</u>	1 (Tony) Invited runner			2 (Sky) Invited runner			3 (Xian)		
<u>Time of Day:</u>	M	A	E	M	A	E	M	A	E
<u>Environment:</u>									
Track	21.11	19.50	25.00	19.45	19.27	23.31	25.55	25.21	27.43
Stone	19.42	21.66	23.63	21.35	20.10	23.76	26.49	25.48	28.12
Grass	20.70	21.96	24.88	21.17	20.35	23.12	27.72	28.05	28.95

In this table, the test subjects in the first column are the three runners and the second column are three time period factors M (morning), A (afternoon), E (evening). In the fourth- sixth column are these three environment factors. It allows us to more

intuitively know the data that the three runners ran out at different times and different places.

Data collection

In order to carry out the design, I invited two other runners. Use runners' names as markers to distinguish each person's data. Which gives us column operators (Tony, Sky, Xian). At this point, the experiment is ready to begin. The experiment was carried out on stadium track, football field and the buffer zone between the spectators and the track of Centennial Stadium in UVic. This corresponded to the racing track, sod and asphalt road. The experiment was done at three time periods, morning, afternoon and evening to test the effect of time on running speed.

The specific experiment procedure was followed:

1. All runners need to eat breakfast before the morning period to make sure that all runners have enough energy to finish the experiment.
2. All runners need to fully warm up and stretch for at least 5 minutes to improve the athletic and flexibility, and also reduce injury from happening.
3. Measure the distance of 100 meters on all three running surfaces (racing track, sod, asphalt road) to ensure the accuracy of the experiment.
4. All runners sprinted on three different surfaces with the same shoes and recorded the time consumption for each runner.
5. For consistency the runner need to take 10 minutes break after each running
6. Repeat the process above for all of the different treatment combination Until the time for each runner run on each type of road surface in each period of time was collected.

The procedure above deviated from the original plan in the method that used to record the time. The original plan step 4 would have been:

4. All the athletes sprinted on three different surfaces and recorded their time consumption, then changed their shoes and repeated the process.

The original plan would have theoretically measured running time more accurately. Because each runner's foot has a different shape the same shoe will have a different fit for different runners. Therefore, the data collection with different runners in different shoes as much as possible to avoid runners being affected by equipment. Unfortunately, the original plan was unrealistic. First, all runners are physically unable to perform dozens of sprints in a short period of time. So at the end of each sprint change the shoes and sprint again is impossible. Second, the experiment can't guarantee that every pair of shoes have the same degree of wear. This also affects the accuracy of the measured data. The third factor is lack of time, because the runners had a lot of exams during the experiment, so the time needed for the experiment must be shortened.

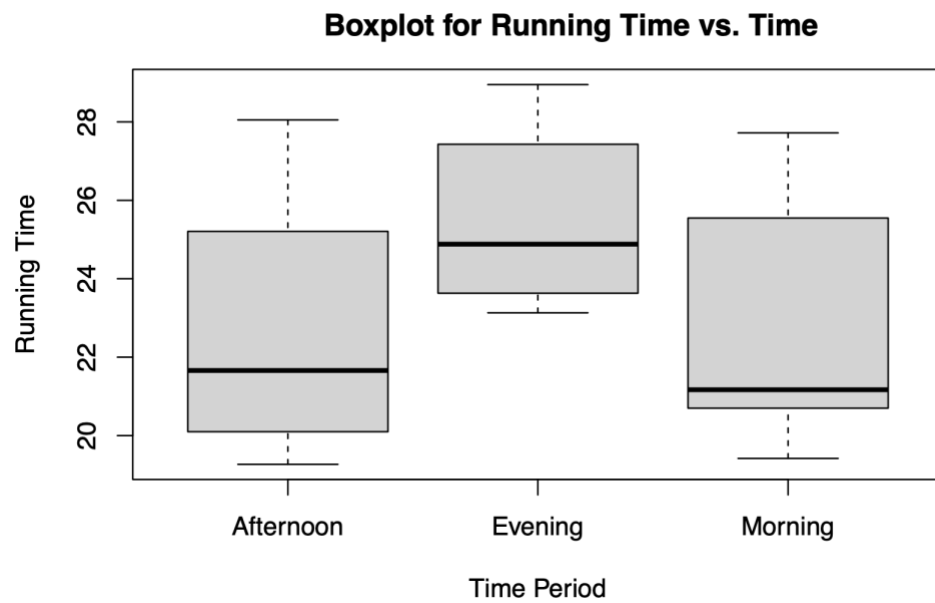
One source of error in the design was human reaction time is not taken into account. Because we don't have professional equipment like flare guns. So runners start the experiment by checking the judge waving his hand. This results in runners not being able to respond immediately in the afternoon when the sun is strong and the lights are dark at night. But the error is so small that are able to be ignored.

Another source of error in the design was the problem of physical decline in running. Because runners are not professional athletes, this results in runners getting slower as they run out of steam in successive tests.

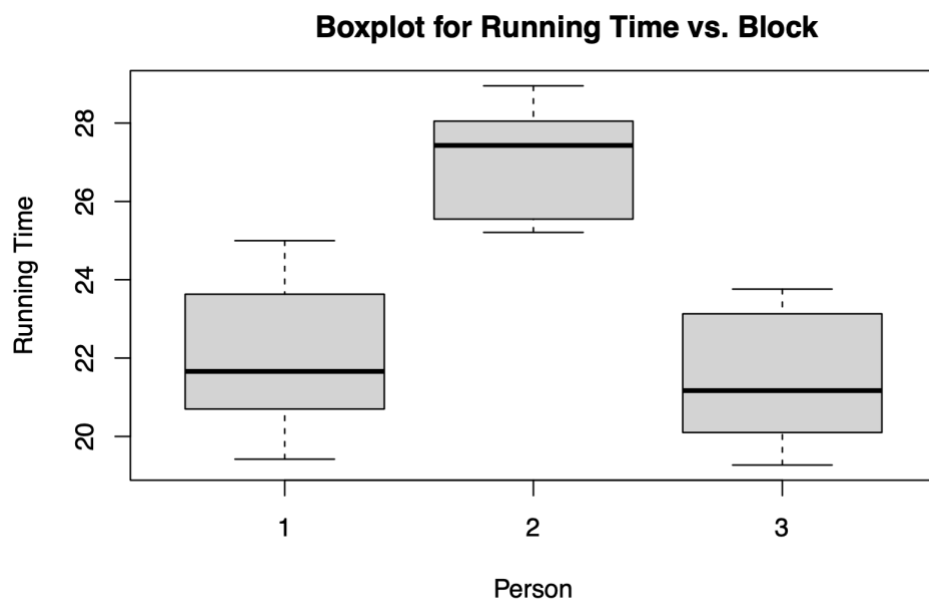
The data collection procedure undoubtedly introduced errors into the design, but overall, the data was representative of the results as seen during the experiment. Due to the error described above, the data does show the effect of time on running speed. It's appropriate to proceed with analysis on the collected data.

Data Analysis:

1. BoxPlot



The distribution of running time on evening are significantly different from morning and afternoon.



2. ANOVA

```
##                               Df Sum Sq Mean Sq F value    Pr(>F)
## factor(Time)                 2   49.95    24.97   27.940 6.03e-06 ***
## factor(Environment)          2    6.96     3.48    3.892  0.042  *
## factor(Block)                2  173.54    86.77   97.072 1.13e-09 ***
## factor(Time):factor(Environment) 4    2.33     0.58    0.651  0.634
## Residuals                   16   14.30     0.89
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Based on the ANOVA table, the time, environment and Block factor are significant but the interaction of time and environment factor are not significant.

Next, non-significant factors are removed, the results are shown below.

```
##                               Df Sum Sq Mean Sq F value    Pr(>F)
## factor(Time)                 2   49.95    24.97  30.035 9.45e-07 ***
## factor(Environment)          2    6.96     3.48    4.184  0.0303  *
## factor(Block)                2  173.54    86.77 104.351 2.62e-11 ***
## Residuals                   20   16.63     0.83
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

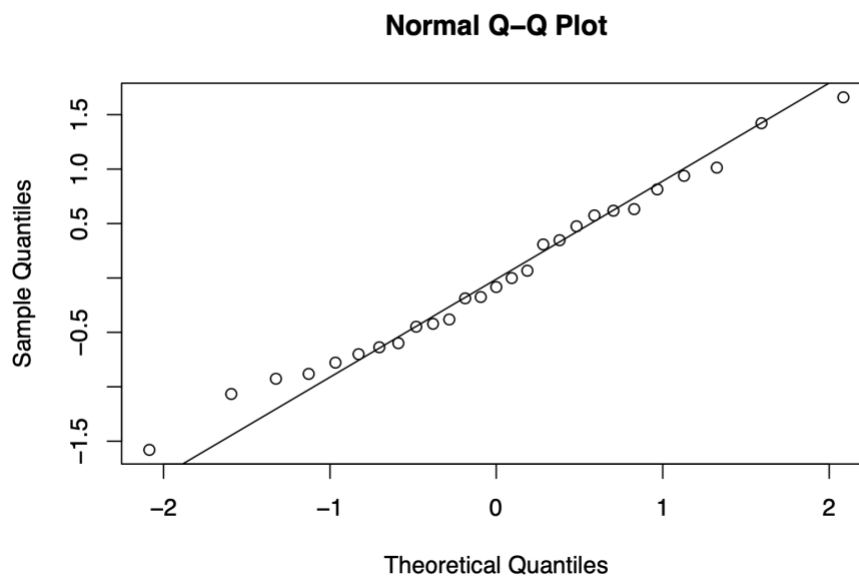
3. Tukey Test

```
## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = RunningTime ~ factor(Time) * factor(Environment) + factor(Block) - factor(Time):f
##
## $`factor(Time)`
##          diff          lwr          upr      p adj
## Evening-Afternoon 2.958889 1.8713525 4.046425 0.0000032
## Morning-Afternoon 0.1533333 -0.9342031 1.240870 0.9325009
## Morning-Evening -2.8055556 -3.8930920 -1.718019 0.0000067
##
## $`factor(Environment)`
##          diff          lwr          upr      p adj
## Stone-Grass -0.7666667 -1.854203 0.3208698 0.2005147
## Track-Grass -1.2311111 -2.318648 -0.1435747 0.0248881
## Track-Stone -0.4644444 -1.551981 0.6230920 0.5367445
##
## $`factor(Block)`
##          diff          lwr          upr      p adj
## 2-1 5.0155556 3.928019 6.1030920 0.0000000
## 3-1 -0.6633333 -1.750870 0.4242031 0.2928616
## 3-2 -5.6788889 -6.766425 -4.5913525 0.0000000
```

For the pairwise comparisons, I assumed $\alpha = 0.05$, there are significant differences between the following pairs, Evening-Afternoon, Morning-Evening and Track-Grass.

4. Model Adequacy Checking

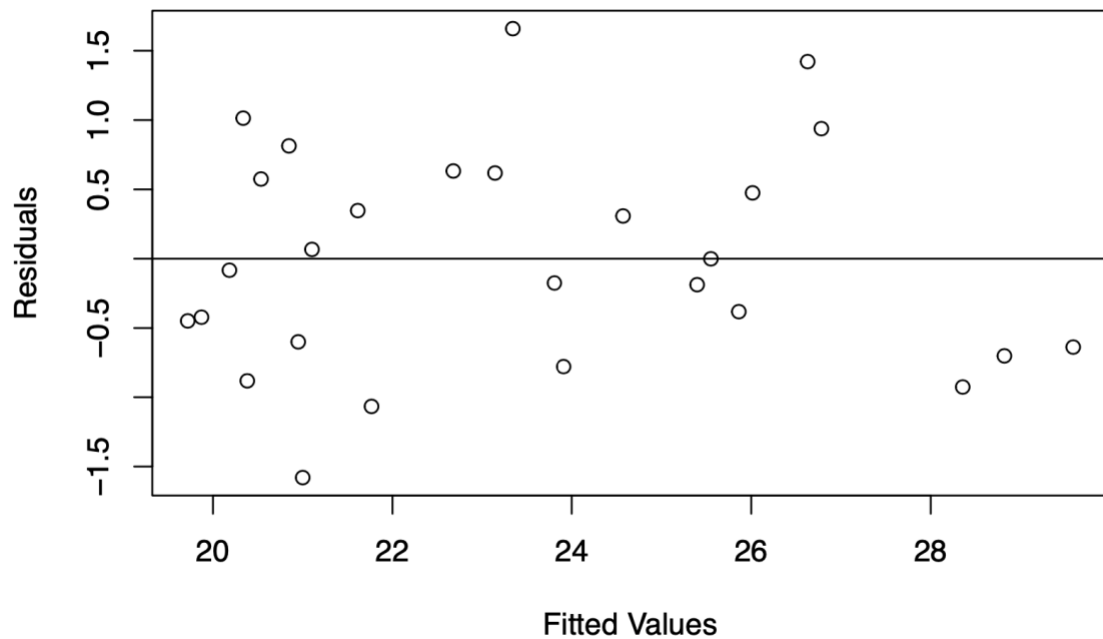
a. Normal Probability Plot & Shapiro Wilk Test




```
##  
## Shapiro-Wilk normality test  
##  
## data: RunningTime.res  
## W = 0.98411, p-value = 0.941
```

Based on the normal qq-plot, most of the points followed the straight line, and since the p-value of shapiro-test is greater than 0.05, the normality assumption is valid.

b. Residuals v.s Fitted values



The points are randomly scattered, so the constant variance assumption is valid.

Conclusion

The goal of designing this experiment is to determine whether the time of day will affect the running speed or not. If the time of day does affect the running speed, what the best running times would be? The three runners were considered as three blocks in this experiment. Factors of time (morning(6-8am), afternoon(12-2pm) and evening(8-10pm)) and environment (track, stone and grass fields) are taken in this experiment.

Based on the data collection and analysis, the time of day will affect the running speed and the best time for running faster is in the morning or afternoon. From the pairwise comparisons, there are significant differences between the factors of evening and afternoon, also the morning and evening.

To do further research in the future, there are some methods which would enhance the experiments. Due to the limited number of class size, only three runners were tested in this experiment. Trying a larger sample size would improve the experiment. Also testing under the different outdoor temperatures such as test during cold or hot morning, afternoon and evening ensures the variability of environments. For the time factor, by testing on different days, it would improve the accuracy and consistency. Due to the lack of budget, runners wore their own outfits and shoes instead of dressing uniformly. For the future, it will be better to test running with the same new shoes for more precise results. Runners could wear the same athletic wear which helps them perform better in running.